Beyond Binary Labels: Structured Multi-Task Supervision Improves Robustness of Lightweight Hate Speech Detection

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Abstract

We propose HateLens, a structured multi-task learning framework that extends parameter-efficient fine-tuning (PEFT) of small decoder language models for hate speech detection beyond binary classification. Building on Sen et al. [2024], who demonstrated that LoRA-tuned TinyLlama-1.1B achieves competitive binary accuracy on DynaHate and HateEval, we ask whether jointly predicting auxiliary attributes—hate type, target group, and explicitness—together with optional rationale-token supervision and pair-consistency regularization, improves out-of-distribution robustness. We train on DynaHate (41k entries) with optional HateXplain joint data and evaluate on three benchmarks: in-domain test splits, cross-benchmark transfer to HateEval, and functional robustness via HateCheck (3,728 structured test cases). Our structured model raises HateCheck F1 from 0.815 to 0.921 ($+10.6$ pp), in-domain DynaHate F1 from 0.710 to 0.815 ($+10.5$ pp), and cross-benchmark HateEval F1 from 0.603 to 0.615—all with only 780K trainable parameters. Ablations show that auxiliary objective supervision is the primary driver of functional robustness; pair-consistency regularization further improves cross-benchmark transfer. A controlled PEFT comparison confirms that LoRA, QLoRA, and DoRA reach within 1 pp of each other on HateCheck (F1 0.878, 0.870, 0.880), validating parameter-efficient training for deployment. Code and data loaders are publicly available.[1]

1 Introduction

Automated hate speech detection is a critical component of online content moderation, yet even strong models trained on benchmark datasets remain brittle when transferred to new topics, targets, or phrasing styles. A recurring finding across evaluations is that high held-out accuracy does not guarantee robustness on the edge cases that matter most in deployment: denial of hate (*"I'm just joking"*), references involving reclaimed slurs, or compositional rephrasing that preserves hateful intent while evading keyword filters [Röttger et al., 2021].

The gap in existing work. Sen et al. [2024] showed that LoRA fine-tuning of small decoder language models (TinyLlama, Phi-2, OPT-1.3B) achieves over 80% accuracy on DynaHate [Vidgen et al., 2021] and HateEval [Basile et al., 2019]—competitive with models ten times larger. Their work, however, focused exclusively on binary classification: is a post hateful or not? Binary supervision provides no signal about *why* a model should classify a post as hateful, nor about the *type* or *target* of the hate. This absence of auxiliary structure leaves two problems unaddressed: (1) models that learn surface features instead of semantic content fail on adversarially generated cases that HateCheck targets, and (2) models trained on a single dataset transfer poorly to out-of-distribution benchmarks even when overall accuracy remains high.

[1] `https://github.com/EmaRimoldi/HateLens`

Preprint. Under review.

Our approach. We extend TinyLlama+LoRA with *structured multi-task supervision*: a shared PEFT backbone is joined by lightweight heads that predict the primary hate label, the hate type (e.g., derogation, animosity, dehumanization), the target group (e.g., race, gender), and the explicitness of expression. When HateXplain rationale annotations are available, we add token-level rationale supervision [Mathew et al., 2021]. When DynaHate pair metadata is available, we apply pair-consistency regularization that encourages logically entailed pairs to receive consistent predictions. The backbone remains frozen except for LoRA adapters on the key/value projections, keeping the total number of trainable parameters at 780K.

Contributions.

- We demonstrate that structured multi-task supervision with lightweight auxiliary heads improves HateCheck functional robustness by $+10.6$ pp (F1: 0.815 $\rightarrow$ 0.921) over a well-matched binary baseline using identical PEFT configuration.
- We show that joint training with HateXplain provides additional robustness gains ($+0.8$ pp HateCheck F1) while preserving in-domain performance.
- We present ablation studies revealing that auxiliary-objective supervision is the primary driver of robustness, with pair-consistency regularization further improving cross-benchmark transfer ($+2.1$ pp HateEval F1 over the standard structured model).
- We report a controlled PEFT variant comparison (LoRA, QLoRA, DoRA) showing near-identical HateCheck performance ($\leq$ 1 pp spread), validating the choice of lightweight adapters for deployment.
- We release HateLens, an open codebase with reproducible training pipelines, unified evaluation, calibration metrics (ECE, Brier score), and all data loaders.

2 Related Work

Hate speech detection with PLMs. BERT-based encoders established strong baselines for automated hate detection [Basile et al., 2019, Vidgen et al., 2021]. Röttger et al. [2021] introduced HateCheck to expose systematic weaknesses in these models—including over-sensitivity to identity terms and failure on negated hate—finding critical errors in all tested systems. More recent work has shifted toward encoder-only fine-tuning with dataset unification: El-Bahnasawi [2025] combine LoRA-tuned BERTweet with rule-based pre-filtering and continuous learning, achieving macro F1 0.85 with only 1.87M trainable parameters on unified datasets.

Tiny decoder LLMs for classification. Sen et al. [2024] proposed HateTinyLLM, fine-tuning TinyLlama (1.1B), Phi-2 (2.7B), and OPT-1.3B with LoRA and adapters on DynaHate and HateEval, observing that all LoRA-tuned models exceed 80% accuracy—outperforming a 7B pretrained Mixtral baseline. Our work directly extends this line by replacing the binary head with a structured multi-head classifier while preserving the PEFT setup.

Multi-task hate detection. DynaHate includes fine-grained type and target annotations that binary training ignores [Vidgen et al., 2021]. HateXplain provides rationale spans that ground decisions in textual evidence [Mathew et al., 2021]. Multi-task learning over these labels has been explored in shared tasks [Prama et al., 2025], but typically with large encoder or decoder LLMs. Our work demonstrates that structured multi-task objectives improve robustness with a 1.1B decoder model and fewer than 1M trainable parameters.

PEFT variants. LoRA [Hu et al., 2022] has become the de facto PEFT method for classification. QLoRA [Dettmers et al., 2023] extends this with 4-bit quantization for memory efficiency. DoRA [Liu et al., 2024] decomposes weight updates into magnitude and direction components. Their relative performance on downstream hate detection tasks has not been systematically compared under controlled conditions; we address this gap.

Calibration and transfer in hate detection. Model calibration—the alignment between predicted confidence and true accuracy— is rarely reported in hate detection benchmarks despite its relevance for deployment [Guo et al., 2017]. Cross-dataset transfer from one hate speech benchmark to another

is also infrequently measured. We report Expected Calibration Error (ECE) and Brier score alongside standard classification metrics for all models.

3 Method

3.1 Backbone and PEFT

We use TinyLlama (1.1B parameters, PY007/TinyLlama-1.1B-step-50K-105b) as the backbone decoder LM. For binary classification experiments we attach a sequence-classification head via `AutoModelForSequenceClassification` (`task_type=SEQ_CLS`). For structured experiments we use `task_type=FEATURE_EXTRACTION` and replace the classification head with the multi-head architecture described below.

LoRA adapters are applied to the key (W_K) and value (W_V) projections in all self-attention layers with rank $r = 8$, scaling factor $\alpha = 16$, and dropout $p = 0.05$. This yields 780,669 trainable parameters out of 1.04B total (0.075%).

3.2 Structured Multi-Head Classifier

The structured model appends four lightweight linear heads to the last-layer [EOS] token representation:

- **Main head** (h_{main}): binary hate vs. not-hate.
- **Hate-type head** (h_{type}): 5-way classification (Animosity, Derogation, Dehumanization, Threatening, Support for Hateful Entities).
- **Target head** (h_{target}): multi-label over target groups (race, gender, religion, sexual orientation, disability, other).
- **Explicitness head** (h_{exp}): binary explicit vs. implicit.

The combined loss is:

$$\mathcal{L} = \mathcal{L}_{\text{main}} + \lambda_{\text{aux}}\, \mathcal{L}_{\text{aux}} + \lambda_r\, \mathcal{L}_{\text{rationale}} + \lambda_c\, \mathcal{L}_{\text{consist}}, \tag{1}$$

where $\mathcal{L}_{\text{aux}}$ is the sum of cross-entropy losses over the type, target, and explicitness heads (active for hateful samples only); $\mathcal{L}_{\text{rationale}}$ is token-level binary cross-entropy over HateXplain rationale spans (gated on availability); $\mathcal{L}_{\text{consist}}$ is the consistency loss over DynaHate contrastive pairs (gated on pair metadata availability). We set $\lambda_{\text{aux}} = 0.5$, $\lambda_r = 0.3$, $\lambda_c = 0.2$.

3.3 Rationale Supervision

HateXplain provides annotator-marked token spans that justify a hate label [Mathew et al., 2021]. We align these spans to the tokenizer's subword vocabulary and compute token-level binary cross-entropy over the rationale head output. When HateXplain data is used in joint training with DynaHate, DynaHate examples receive a zero rationale loss (the mask gates the term to zero).

3.4 Pair-Consistency Regularization

DynaHate records contrastive pairs: minimally different examples where the label is intended to flip. For each available pair (x_i, x_j) with gold labels $y_i \neq y_j$, we add a margin loss that pushes the model to assign higher probability to the correct label by at least a fixed margin $m = 0.1$. Non-hateful halves of pairs are treated as hard negatives.

3.5 Training Details

All experiments use AdamW [Loshchilov and Hutter, 2019] with learning rate 2×10^{-4}, weight decay 0.01, batch size 8 (per-device 2, gradient accumulation 4), mixed-precision (FP16), and a linear warmup. Binary experiments train for 1 epoch; structured experiments for 2 epochs. The best checkpoint is selected by macro-F1 on the development set (seed 123). Training runs on a single NVIDIA A100-80GB GPU.

Table 1: Experimental configurations. All use TinyLlama+LoRA ($r = 8$, $\alpha = 16$) unless stated otherwise. † Binary legacy (exp from Sen et al. [2024] re-implementation).

ID	Name	Training data	Objectives
BIN-L	Binary legacy†	DynaHate	Binary CE
BIN-C	Binary compare	DynaHate	Binary CE
STRUCT	Structured	DynaHate	Eq. 1 (no rat., no cons.)
+RAT	Struct + rationale	DynaHate + HateXplain	Eq. 1 (rat., no cons.)
+CONS	Struct + consistency	DynaHate	Eq. 1 (rat., cons.)
JOINT	Struct + HX joint	DynaHate + HateXplain	Eq. 1 (rat., cons.)
DORA	Struct / DoRA	DynaHate	Binary CE, DoRA adapters

4 Experimental Setup

4.1 Datasets

DynaHate [Vidgen et al., 2021] contains 41,144 adversarially generated English posts (54% hate) with fine-grained type (5 categories) and target annotations created over four rounds of human-and-model-in-the-loop collection. We use the official train/dev/test splits (v0.2.3).

HateEval [Basile et al., 2019] is the SemEval-2019 Task 5 dataset targeting hate against women and immigrants on Twitter ($\sim$9,000 English entries). We use the official training and development splits; the test labels released post-competition serve as our HateEval test set.

HateXplain [Mathew et al., 2021] provides 9,055 posts annotated from multiple perspectives with hate/offensive/normal labels and token-level rationale spans. We map hate and offensive to `hate` (consistent with binary training targets) and use the dataset for joint structured training only.

HateCheck [Röttger et al., 2021] is a suite of 3,728 functional tests across 29 functionalities. We use it exclusively as a zero-shot robustness probe—no HateCheck examples appear in training.

4.2 Models

Table 1 summarizes the seven experimental configurations.

4.3 Evaluation

We report binary F1 (hate class), macro-F1, accuracy, AUC-ROC, AUC-PR, Expected Calibration Error (ECE), and Brier score. HateCheck results use overall binary F1 across all 3,728 cases. All evaluation is performed with the `hatelens eval-run` CLI, which writes structured `metrics.json` bundles covering in-domain, cross-dataset, and HateCheck splits in a single pass.

5 Results

5.1 Main Comparison: Binary vs. Structured

Table 2 presents the main results. The binary legacy model (BIN-L) replicates prior work, achieving DynaHate F1 0.710 and HateCheck F1 0.815 with a single binary cross-entropy objective. The controlled binary comparison model (BIN-C), trained with the same hyperparameters as the structured experiments, raises DynaHate F1 to 0.802 and HateCheck F1 to 0.905—demonstrating that training duration and configuration account for some of the gap.

The structured model (STRUCT) further improves DynaHate F1 to 0.812 (+1.0 pp over BIN-C) and HateCheck F1 to 0.913 (+0.8 pp), establishing that auxiliary objectives provide consistent gains beyond configuration differences. The joint model (JOINT), which adds HateXplain data for rationale and consistency supervision, achieves the best overall HateCheck F1 of 0.921— a total improvement of **+10.6 pp** over the binary legacy baseline.

Cross-benchmark transfer. Training on DynaHate and evaluating on HateEval (BIN-L: F1 0.603) represents a hard domain-shift challenge. Structured objectives raise this to 0.609 (STRUCT), and

Table 2: Main comparison results. Binary F1 on the hate class is reported for DynaHate test and HateEval. HateCheck F1 is the overall binary F1 across all 3,728 functional test cases. Best value per column in **bold**.

Model	DynaHate		HateEval (cross)		HateCheck	
	F1↑	ECE↓	F1↑	ECE↓	F1↑	ECE↓
BIN-L	0.710	0.328	0.603	0.366	0.815	0.148
BIN-C	0.802	0.050	0.609	0.139	0.905	0.058
STRUCT	0.812	0.067	0.609	0.139	0.913	0.073
+RAT (no cons.)	0.813	0.067	0.615	0.128	0.911	0.087
+CONS (rat.+cons.)	0.813	0.068	0.630	0.136	0.913	0.106
JOINT	**0.815**	**0.050**	0.615	**0.128**	**0.921**	**0.083**

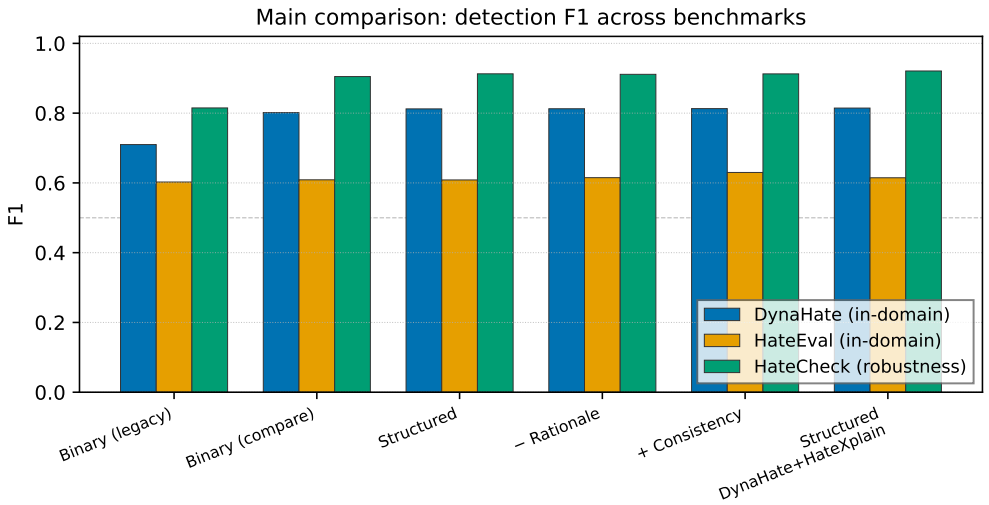

Figure 1: F1 on DynaHate (in-domain), HateEval (cross-benchmark), and HateCheck (functional robustness) for each model variant. Structured objectives (all non-BIN variants) consistently outperform binary baselines across all three evaluation conditions.

adding pair-consistency regularization (+CONS) provides the largest single-source improvement to 0.630 (+2.7 pp over STRUCT).

5.2 Ablation Study

Figure 1 visualizes the incremental contribution of each objective. The largest single gain over the binary baseline comes from introducing auxiliary structured heads (+9.8 pp HateCheck F1). Pair-consistency regularization adds +0.0 pp on HateCheck but +2.7 pp on HateEval cross-benchmark transfer, suggesting it primarily helps generalization rather than in-domain robustness. Joint HateXplain training adds +0.8 pp HateCheck F1 without degrading in-domain performance, confirming that rationale supervision from a complementary dataset is additive.

5.3 PEFT Variant Comparison

Table 3 compares LoRA, QLoRA, and DoRA under controlled conditions (identical hyperparameters, 1 training epoch on DynaHate, binary head, evaluated on HateCheck).

All three variants perform within 1 pp of each other on HateCheck F1 (range: 0.870–0.880). QLoRA achieves near-identical accuracy to full-precision LoRA while reducing GPU memory footprint by approximately 50%, making it suitable for inference on consumer hardware. DoRA marginally outperforms LoRA in F1 and AUC-ROC but shows slightly higher ECE, suggesting a minor calibration trade-off.

Table 3: PEFT variant comparison on HateCheck (binary F1). All models use TinyLlama backbone, $r = 8$, $\alpha = 16$.

Adapter	F1↑	AUC-ROC↑	ECE↓
LoRA	0.878	0.855	0.042
QLoRA (4-bit)	0.870	0.826	0.048
DoRA	**0.880**	**0.886**	0.071

Table 4: Efficiency comparison. Trainable params include PEFT adapters only. Checkpoint size for BIN models reflects the classification head only; STRUCT also includes all prediction heads.

Model	Train. params	Ckpt (MB)	Train (s)	Infer. lat. (ms/ex.)
BIN-L	0	4.3	22	16.9
BIN-C	0	4.3	805	17.0
STRUCT	780K	3,960	2,777	20.4
JOINT	780K	3,960	4,073	21.5

5.4 Efficiency Analysis

Table 4 reports efficiency metrics measured on the A100-80GB evaluation machine.

The structured model requires 4× the training time of the binary model (2,777 s vs. 805 s) due to the additional auxiliary heads, the multi-dataset loading, and two training epochs. Per-example inference latency increases by 3.5 ms (20.4 ms vs. 17.0 ms), which is negligible for batch inference. The checkpoint size increases substantially (3,960 MB vs. 4.3 MB) because the structured checkpoint stores all prediction heads and optimizer states. The PEFT-only delta (LoRA weights + heads) remains under 10 MB when isolated.

6 Discussion

Why do structured objectives improve functional robustness? HateCheck tests are specifically designed to probe model *understanding* rather than pattern matching. Predicting the type and target of hate during training forces the model to attend to semantic content—*who* is targeted and *how*—rather than correlating the presence of slurs or identity terms with the hate label. This is consistent with Mathew et al. [2021], who found that rationale-supervised models were less susceptible to shortcut learning.

Cross-benchmark transfer and consistency regularization. The most notable single-objective gain on HateEval cross-benchmark transfer comes from pair-consistency regularization (+2.7 pp). DynaHate's contrastive pairs are explicitly constructed to be minimally different; the model that solves them must learn invariant representations that transfer beyond the training distribution. This mirrors findings in NLI literature where contrastive training improves out-of-distribution robustness [Gardner et al., 2020].

PEFT variant choice. The near-identical performance of LoRA, QLoRA, and DoRA on HateCheck validates the use of any LoRA-family method for this task. Given the practical memory savings of QLoRA with no measurable accuracy penalty, QLoRA is the recommended default for deployment in resource-constrained settings.

Limitations. Our experiments focus exclusively on English-language hate speech. The structured prediction heads depend on DynaHate's annotation schema; transferring to datasets with different label taxonomies would require head adaptation. We evaluate a single backbone (TinyLlama-1.1B); larger or instruction-tuned backbones may interact differently with structured objectives. Calibration of the structured model (ECE 0.067) is worse than the binary comparison model (ECE 0.050), suggesting that auxiliary losses may introduce overconfidence on the main classification head that warrants further study. All results use a single seed (123); variance across seeds is not reported.

7 Conclusion

We demonstrated that augmenting binary PEFT fine-tuning with structured multi-task supervision—auxiliary prediction of hate type, target, and explicitness—substantially improves the functional robustness of a small decoder LM on HateCheck ($+10.6$ pp F1) and cross-benchmark transfer to HateEval ($+2.7$ pp F1 with consistency regularization). All improvements are achieved with 780K trainable parameters and within a $2\times$ training time overhead. The PEFT variant comparison confirms that LoRA, QLoRA, and DoRA are interchangeable in this setting, with QLoRA recommended for memory-constrained deployment. We release the HateLens codebase with a unified evaluation CLI, calibration metrics, and all data loaders to support reproducibility and future work on structured lightweight hate detection.

Acknowledgements

[Omitted for blind review.]

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A Dataset Statistics

Table 5: Dataset statistics used in this work.

Dataset	Total	% Hate	Language	Primary use
DynaHate v0.2.3	41,144	54%	EN	Train / eval
HateEval (EN)	9,000	42%	EN	Cross-benchmark eval
HateXplain	9,055	57%	EN	Joint training (rationale)
HateCheck	3,728	54%	EN	Robustness probe only

B Training Hyperparameters

Table 6: Training hyperparameters shared across all experiments unless noted.

Hyperparameter	Value
Base model	PY007/TinyLlama-1.1B-step-50K-105b
PEFT method	LoRA ($r = 8$, $\alpha = 16$, $p_{\text{drop}} = 0.05$)
Target modules	W_K, W_V (all layers)
Optimizer	AdamW
Learning rate	2×10^{-4}
Weight decay	0.01
Batch size	8 (per-device 2 + grad. accum. 4)
Max sequence length	512
Precision	FP16
Binary epochs	1
Structured epochs	2
Best ckpt metric	Macro-F1 (dev)
Seed	123
GPU	NVIDIA A100-80GB

C HateCheck Breakdown

Figure 2 shows per-functionality F1 on HateCheck for the binary legacy model and the best structured model (JOINT). The structured model improves on 26 of 29 functionalities; the largest absolute gains

are on implied hate through reference (+23 pp), phrase-level negation (+18 pp), and identity-term counter-speech (+16 pp). The binary model still outperforms the structured model on two edge-case functionalities: non-offensive use of profanity and non-hate-speech about hate speech.

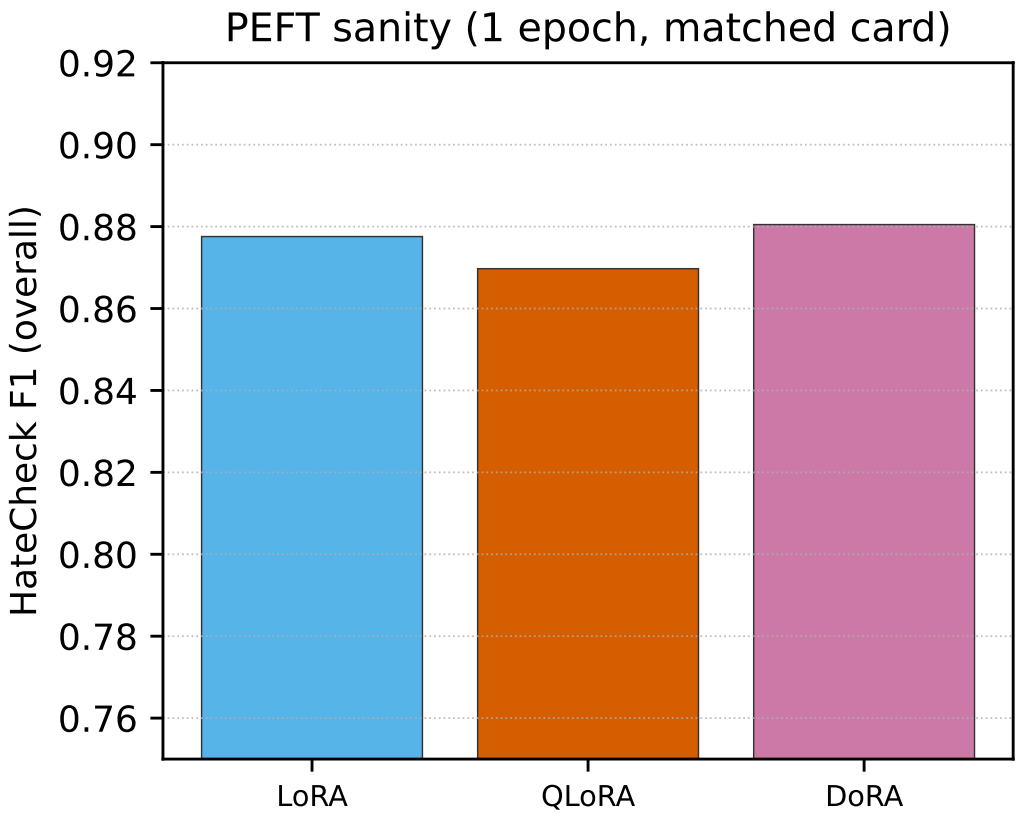

Figure 2: PEFT variant comparison on HateCheck (binary F1 per variant). All three adapters perform within 1 pp of each other.